SECURITE IOT

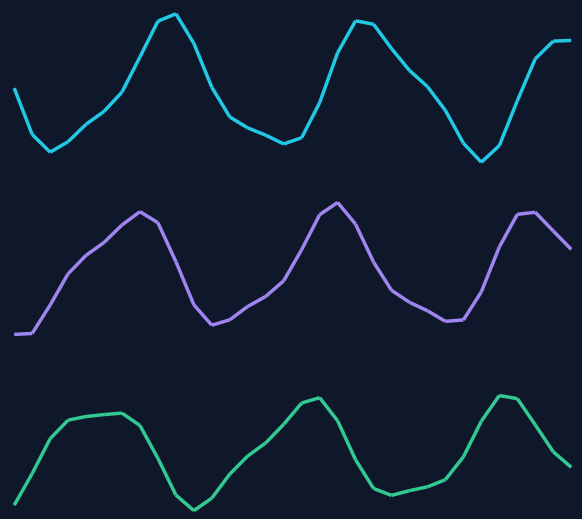
PRIVACY

# PrivIMU

## Motion Anonymity Attack

Can anonymous accelerometer + gyroscope tra[illegible]ntity?

**ML attacker**




Deliverables: repo + Colab + Streamlit + slides + report + metrics.json

# Pourquoi les IMU sont un enjeu IoT

Les capteurs de mouvement sont discrets, permanents et souvent consideres comme peu sensibles.

**phone**

Smartphone

**watch**

Wearable

**heart**

Health app

**run**

Sport app

**map**

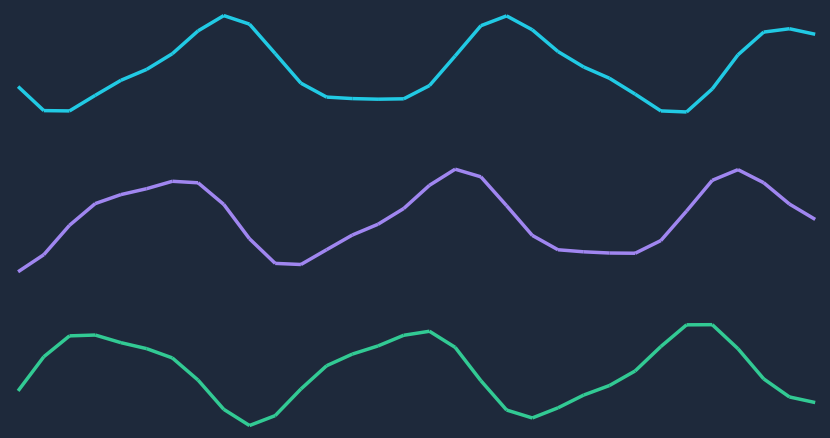
Mobility

**Problem: these traces are often shared as "anonymous" telemetry.**

# Question scientifique

Anonyme ne veut pas dire non re-identifiable.

## Signal IMU



No name. No email. No face.

## Identity posterior

Hypothesis: motion traces act as behavioral quasi-identifiers.

# Threat model

Un attaquant modeste, pas un laboratoire militaire.

**1**

### Enrollment data

Traces publiques ou historiques pour apprendre les signatures.

→

**2**

### Anonymous IMU CSV

Acceleration + gyroscope-derived signal, no explicit ID.

→

**3**

### Classifier

Random Forest or 1D-CNN, CPU-friendly.

→

**4**

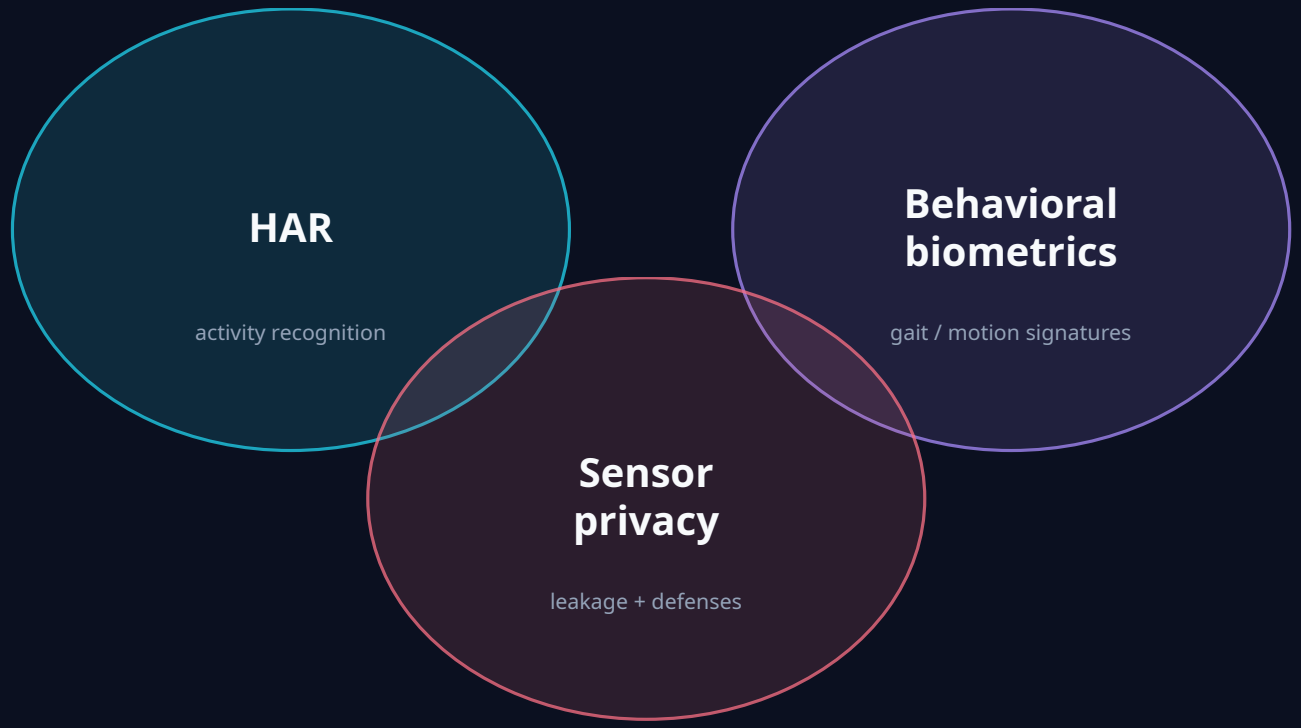
### Top-3 identity

Posterior over 24 enrolled subjects.

Privacy framing: this is an academic defensive demo, using a public dataset.

# Positionnement scientifique

PrivIMU relie HAR, biometrie comportementale et privacy IoT.




## Our gap

Closed-set identity attack on MotionSense
Reproducible pipeline, not only slides
Live demo with privacy entropy
Mitigation discussion: noise, minimization, edge, DP

# Dataset: MotionSense

Public IMU dataset collected from smartphone motion sensors.

| 24 | 6 | 50 Hz | iPhone | pocket |
|---|---|---|---|---|
| data subjects | activities | sampling | device | placement |

## Activities

- walking
- jogging
- upstairs
- downstairs
- sitting
- standing

## Channels used by PrivIMU

rotationRate.x/y/z
userAcceleration.x/y/z
1-second windows with 50% overlap

# Pipeline reproductible

Chaque chiffre doit venir du code et de reports/metrics.json.

**download.py**
MotionSense ZIP

→

**data.py**
CSV -> arrays

→

**features.py**
windows + features

→

**train.py**
RF / CNN

→

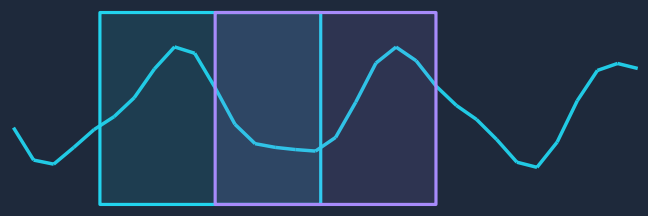
**evaluate.py**
metrics + gates

**Rule: no manual metric in slides or report. Source of truth = reports/metrics.json**

# Preprocessing and features

From raw time-series to identity-classification features.

## Windowing



50 samples
= 1s

50% overlap

## Feature set

mean / std / min / max
RMS and range
skewness / kurtosis
spectral entropy
energy-like descriptors

## Models

Random Forest baseline
1D-CNN extension
Top-1 / Top-3 metrics
latency per window

# Evaluation protocol

Avoiding leakage is as important as the classifier.

## Split strategy

Group-based split by activity/trial prevents adjacent windows from the same trial leaking across train/test.

**Bad split** window 1 | window 2 | window 3 mixed randomly

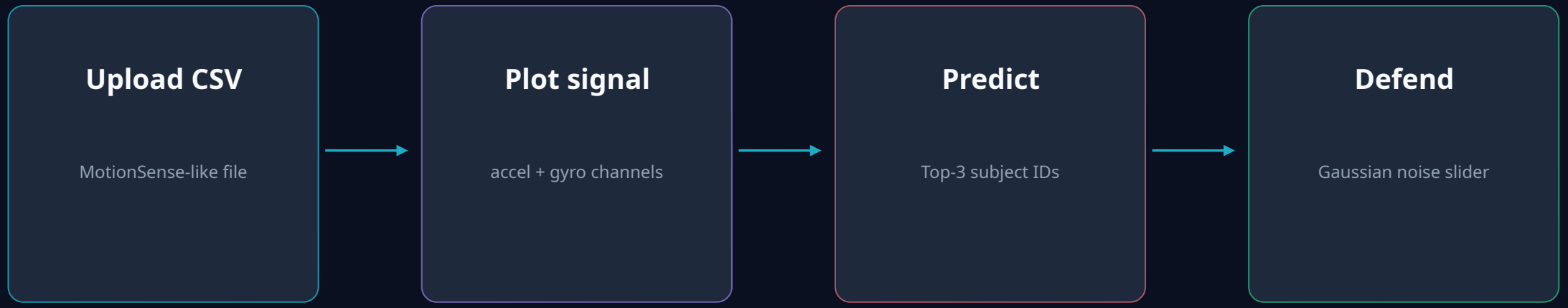
**Good split** whole trial held out as evaluation group

## Metrics

Top-1 accuracy
Top-3 accuracy
Macro-F1
Confusion matrix 24x24
Per-subject F1
Privacy entropy leakage
Latency / window

# Demo architecture

The demo turns privacy leakage into something visible.

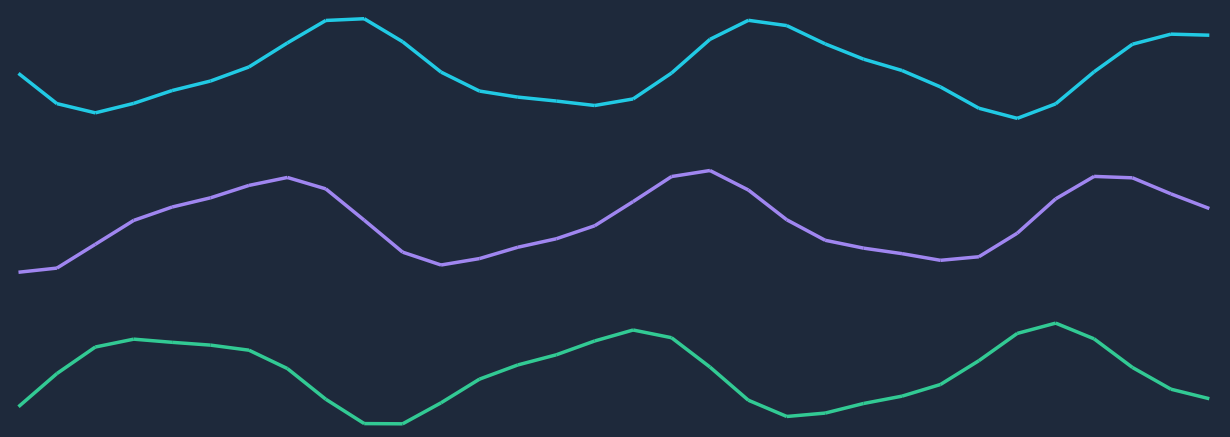



Fallback mode keeps the UI clickable before training; real predictions use models/rf.joblib.

# Live attack: from signal to Top-3 identity

The signal has no explicit identity field.

## Anonymous IMU signal



rotationRate.x/y/z + userAcceleration.x/y/z

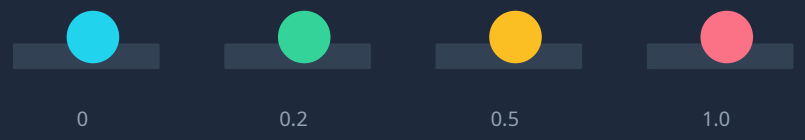
## Top-3 posterior

Replace example bars with live app output during the presentation.

# Defense demo: perturbation vs leakage

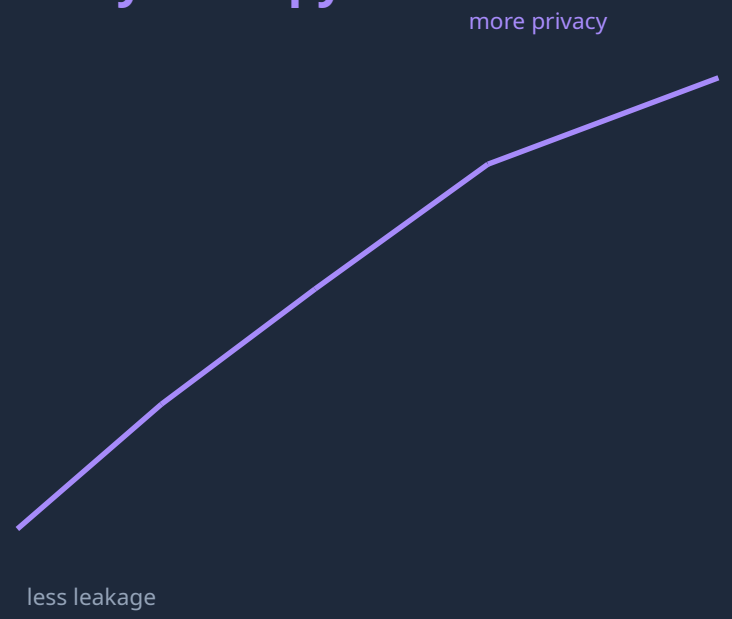
A simple noise slider turns privacy into a measurable trade-off.

## Gaussian noise slider




As sigma increases, identity confidence should drop and posterior entropy should rise.

## Privacy entropy



# Results - generated, not invented

This slide is refreshed from reports/metrics.json after training.

**run train-rf**
Top-1 accuracy

**run train-rf**
Top-3 accuracy

**run train-rf**
Macro-F1

**run train-rf**
Latency / window

## Evidence package

reports/metrics.json = source of truth
reports/confusion_matrix.png
reports/per_subject_f1.png
reports/privacy_entropy_curve.png
commit message includes Verified-By

**No manual metric**

# Limits and mitigations

A strong project is honest about its assumptions.

## Limits

Closed-set identification among enrolled subjects
MotionSense has 24 subjects
Phone-in-pocket setting
Dataset differs from every real deployment
Noise defense is pedagogical, not a formal guarantee

## Mitigations

Data minimization
On-device inference
Purpose limitation
Federated learning
Differential privacy for time-series
Auditable retention policy

# Takeaway

## Anonymous motion data can still be privacy-sensitive.

PrivIMU is delivered as a reproducible mini-lab: repo, Colab, Streamlit demo, report, slides, tests and generated metrics.

GitHub repo
github.com/akiroussama/PrivIMU

Google Colab
reproduce results

Streamlit demo
privimu.streamlit.app